

Title: Sales Data Analysis Using MySQL (ClassicModels)

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Tools: MySQL

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Project Overview

This project focuses on analyzing sales, customer, and order data using MySQL and the ClassicModels database. The objective of this project is to extract meaningful business insights such as total revenue, monthly sales trends, top customers, best-selling products, and country-wise sales performance. The analysis helps understand customer behavior, product performance, and overall business sales trends.

Database Used

Database: ClassicModels

Tables Used:

- Customers – Contains customer information
- Orders – Contains order details
- OrderDetails – Contains product order quantity and price
- Payments – Contains customer payment information
- Products – Contains product information

Tools Used

- MySQL – For writing SQL queries

Problem Statements

1. Find the total number of customers in the database.
2. Show all customers from the USA.
3. Find the top 5 customers with the highest credit limit.
4. Display the number of customers in each country.
5. Find customers whose credit limit is greater than 50,000.
6. Find the total number of orders placed.
7. Show the number of orders placed by each customer.
8. Find the latest order date in the dataset.
9. Display customers who have not placed any orders.

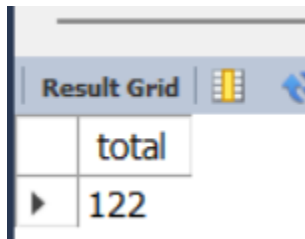
10. Find the top 5 customers with the highest number of orders.
11. Calculate the total revenue generated.
12. Find the total sales for each product.
13. Find the top 10 best-selling products.
14. Calculate total sales by product line.
15. Find the average product price.
16. Display customer name and order number.
17. Show product name and quantity ordered.
18. Find total sales by each customer.
19. Display customer name, order number, and product ordered.
20. Find customers who made payments greater than 100,000.
21. Find the customer who generated the highest revenue.
22. Find the most ordered product.
23. Find the country generating the highest sales.
24. Find customers with payments greater than the average payment.
25. Analyze monthly sales trend.

SQL Queries

1. Find the total number of customers in the database.

```
select count(*) as total_customers  
from customers;
```

output:



Result Grid	
	total
▶	122

2. Show all customers from USA.

```
select * from customers  
where country = 'USA';
```

output:

customerNumber	customerName	Lastname	Firstname	phone	addressLine1	addressLine2	city	state	postalCode	country
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.		Las Vegas	NV	83030	USA
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	5677 Strong St.		San Rafael	CA	97562	USA
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street		San Francis...	CA	94217	USA
131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue		NYC	NY	10022	USA
151	Muscle Machine Inc.	Young	Jeff	2125557413	4092 Furth Circle	Suite 400	NYC	NY	10022	USA
157	Diecast Classics Inc.	Leong	Kelvin	2155551555	7586 Pompton St.		Allentown	PA	70267	USA
161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle		Burlingame	CA	94217	USA
168	American Souvenirs Inc.	Franco	Keith	2035557845	149 Spinnaker Dr.	Suite 101	New Haven	CT	97823	USA
173	Cambridge Collectables Co.	Tseng	Jerry	6175555555	4658 Baden Av.		Cambridge	MA	51247	USA
175	Gift Depot Inc.	King	Julie	2035552570	25593 South Bay Ln.		Bridgewater	CT	97562	USA
181	Vitachrome Inc.	Frick	Michael	2125551500	2678 Kingston Rd.	Suite 101	NYC	NY	10022	USA
198	Auto-Moto Classics Inc.	Taylor	Leslie	6175558428	16780 Pomoton St.		Brickhaven	MA	58339	USA

3.Find the top 5 customers with the highest credit limit.

select * from customers

order by creditlimit desc

limit 5;

output:

customerNumber	customerName	Lastname	Firstname	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmp
141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	C/ Moralzarzal, 86		Madrid		28034	Spain	1370
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	5677 Strong St.		San Rafael	CA	97562	USA	1165
298	Vida Sport, Ltd	Holz	Mihael	0897-034555	Grenzacherweg 237		Genève		1203	Switzerland	1702
151	Muscle Machine Inc	Young	Jeff	2125557413	4092 Furth Circle	Suite 400	NYC	NY	10022	USA	1286
187	AV Stores, Co.	Ashworth	Rachel	(171) 555-1555	Fauntleroy Circus		Manchester		EC2 5NT	UK	1501

4.Display the number of customers in each country.

select country,count(customernumber) as total_customers

from customers

group by country;

output:

Result Grid   Filter Rows:		
	country	total_custome
▶	France	12
	USA	36
	Australia	5
	Norway	1
	Poland	1
	Germany	13
	Spain	7
	Sweden	2
	Denmark	2
	Singapore	3
	Norway	2
	Portugal	2

5.Find customers whose credit limit is greater than 50,000.

select customernumber,city,state,country,creditlimit from customers

where creditlimit > 50000;

output:

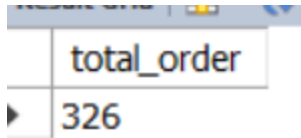
customernumber	city	state	country	creditlimit
112	Las Vegas	NV	USA	71800.00
114	Melbourne	Victoria	Australia	117300.00
119	Nantes	NULL	France	118200.00
121	Stavern	NULL	Norway	81700.00
124	San Rafael	CA	USA	USA 0500.00
128	Frankfurt	NULL	Germany	59700.00
129	San Francis...	CA	USA	64600.00
131	NYC	NY	USA	114900.00
141	Madrid	NULL	Spain	227600.00
144	Luleå	NULL	Sweden	53100.00
145	Kobenhavn	NULL	Denmark	83400.00
146	Lyon	NULL	France	123900.00
148	Singapore	NULL	Singapore	103800.00
151	NYC	NY	USA	138500.00
157	Allentown	PA	USA	100600.00
161	Burlingame	CA	USA	84600.00
166	Singapore	NULL	Singapore	97900.00
167	Bergen	NULL	Norway	96800.00
171	Lille	NULL	France	82900.00
172	Paris	NULL	France	84300.00
175	Bridgewater	CT	USA	84300.00
177	Kita-ku	Osaka	Japan	81200.00

6.Find the total number of orders placed.

select count(ordernumber) as total_orders

from orders;

output:



total_order	
▶	326

7. Show the number of orders placed by each customer.

```
select customernumber,  
count(ordernumber) as total_orders  
from orders  
group by customernumber;
```

output:

	customernumber	total
▶	103	3
	112	3
	114	5
	119	4
	121	4
	124	17
	128	4
	129	3
	131	4
	141	26
	144	4
	145	5
	146	3
	148	5
	151	4
	157	4
	161	4
	166	4
	167	3
	171	2
	172	3
	173	2

8.Find the latest order date in the dataset.

```
select max(orderdate) as latest_order
from orders;
```

output:

latest_order
2005-05-31

9.Display customers who have not placed any orders.

select customernumber from customers

left join orders

using(customernumber)

where ordernumber is null;

output:

	customernumber
▶	125
	168
	169
	206
	223
	237
	247
	273
	293
	303
	307
	335
	348
	356
	361
	369
	376
	409
	443
	459
	465
	477

10. Find the top 5 customers with the highest number of orders.

```
select customernumber, count(ordernumber) as total_orders from customers
```

```
inner join orders
```

```
using(customernumber)
```

```
group by customernumber
```

```
order by total_orders desc
```

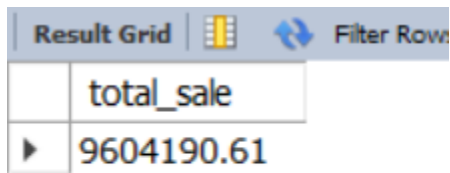
```
limit 5;
```

output:

11. Calculate the total revenue generated.

```
select sum(quantityordered * priceeach) as total_sales  
from orderdetails;
```

output:



Result Grid		Filter Rows
	total_sale	
▶	9604190.61	

12. Find the total sales for each product.

```
select productname, sum(quantityordered * priceeach) as total_sales  
from orderdetails  
inner join products  
using(productcode)  
group by productname;
```

output:

productname	total_sale
1969 Harley Davidson Ultimate Chop...	90157.77
1952 Alpine Renault 1300	190017.96
1996 Moto Guzzi 1100i	109998.82
2003 Harley-Davidson Eagle Drag Bike	170686.00
1972 Alfa Romeo GTA	127924.32
1962 LanciaA Delta 16V	123123.01
1968 Ford Mustang	161531.48
2001 Ferrari Enzo	190755.86
1958 Setra Bus	119085.25
2002 Suzuki XREO	135767.03
1969 Corvair Monza	132363.79
1968 Dodge Charger	98718.76
1969 Ford Falcon	152543.02
1970 Plymouth Hemi Cuda	63489.95
1957 Chevy Pickup	109946.21
1969 Dodge Charger	104210.62
1940 Ford Pickup Truck	105563.12
1993 Mazda RX-7	114648.12
1937 Lincoln Berline	102563.52
1936 Mercedes-Benz 500K Special Ro...	46078.29
1965 Aston Martin DB5	101778.13
1980s Black Hawk Helicopter	144959.91

13. Find the top 10 best-selling products.

```

select productname, sum(quantityordered * priceeach) as total_sales
from products
inner join orderdetails
using(productcode)
group by productname
order by total_sales desc

```

limit 10;

output:

productname	total_sale
1992 Ferrari 360 Spider red	276839.98
2001 Ferrari Enzo	190755.86
1952 Alpine Renault 1300	190017.96
2003 Harley-Davidson Eagle Drag Bike	170686.00
1968 Ford Mustang	161531.48
1969 Ford Falcon	152543.02
1980s Black Hawk Helicopter	144959.91
1998 Chrysler Plymouth Prowler	142530.63
1917 Grand Touring Sedan	140535.60
2002 Suzuki XREO	135767.03

14.Calculate total sales by product line.

select productline,sum(quantityordered * priceeach) as total_sales

from orderdetails

inner join products

using(productcode)

group by productline;

output:

productline	total_sale
Classic Cars	3853922.49
Motorcycles	1121426.12
Planes	954637.54
Ships	663998.34
Trains	188532.92
Trucks and Buses	1024113.57
Vintage Cars	1797559.63

15.Find the average product price.

select avg(buyprice) as avgprice from products;

output:

avgprice
54.395182

16.Display customer name and order number.

select customername,ordernumber from customers

inner join orders

using(customernumber);

output:

customername	ordernumber
Atelier graphique	10123
Atelier graphique	10298
Atelier graphique	10345
Signal Gift Stores	10124
Signal Gift Stores	10278
Signal Gift Stores	10346
Australian Collectors, Co.	10120
Australian Collectors, Co.	10125
Australian Collectors, Co.	10223
Australian Collectors, Co.	10342
Australian Collectors, Co.	10347
La Rochelle Gifts	10275
La Rochelle Gifts	10315
La Rochelle Gifts	10375
La Rochelle Gifts	10425
Baane Mini Imports	10103
Baane Mini Imports	10158
Baane Mini Imports	10309
Baane Mini Imports	10325
Mini Gifts Distributors Ltd.	10113
Mini Gifts Distributors Ltd.	10135
Mini Gifts Distributors Ltd.	10142

17.Show product name and quantity ordered.

select productname,quantityordered from orderdetails

join products

using(productcode);

output:

productname	quantityordered
1969 Harley Davidson Ultimate Chop...	30
1969 Harley Davidson Ultimate Chop...	34
1969 Harley Davidson Ultimate Chop...	41
1969 Harley Davidson Ultimate Chop...	45
1969 Harley Davidson Ultimate Chop...	49
1969 Harley Davidson Ultimate Chop...	36
1969 Harley Davidson Ultimate Chop...	29
1969 Harley Davidson Ultimate Chop...	48
1969 Harley Davidson Ultimate Chop...	22
1969 Harley Davidson Ultimate Chop...	41
1969 Harley Davidson Ultimate Chop...	37
1969 Harley Davidson Ultimate Chop...	23
1969 Harley Davidson Ultimate Chop...	59
1969 Harley Davidson Ultimate Chop...	34
1969 Harley Davidson Ultimate Chop...	45
1969 Harley Davidson Ultimate Chop...	36
1969 Harley Davidson Ultimate Chop...	23
1969 Harley Davidson Ultimate Chop...	41
1969 Harley Davidson Ultimate Chop...	46
1969 Harley Davidson Ultimate Chop...	42
1969 Harley Davidson Ultimate Chop...	41
1969 Harley Davidson Ultimate Chop...	42

18. Find total sales by each customer.

```

select customername, sum(quantityordered * priceeach) as total_sales
from customers
inner join orders
using(customernumber)
inner join orderdetails
using (ordernumber)

```

group by customername

order by total_sales desc;

output:

customername	total_sale
Euro+ Shopping Channel	820689.54
Mini Gifts Distributors Ltd.	591827.34
Australian Collectors, Co.	180585.07
Muscle Machine Inc	177913.95
La Rochelle Gifts	158573.12
Dragon Souvenirs, Ltd.	156251.03
Down Under Souvenirs, Inc	154622.08
Land of Toys Inc.	149085.15
AV Stores, Co.	148410.09
The Sharp Gifts Warehouse	143536.27
Salzburg Collectables	137480.07
Kelly's Gift Shop	137460.79
Anna's Decorations, Ltd	137034.22
Souvenirs And Things Co.	133907.12
Corporate Gift Ideas Co.	132340.78
Saveley & Henriot, Co.	130305.35
Danish Wholesale Imports	129085.12
Rovelli Gifts	127529.69
Reims Collectables	126983.19
L'ordine Souvenirs	125505.57
Scandinavian Gift Ideas	120943.53
Online Diecast Creations Co.	116449.29

19.Display customer name, order number, and product ordered.

select customername,ordernumber,productname

from customers

inner join orders

using(customernumber)

inner join orderdetails

using(ordernumber)

inner join products

using(productcode);

output:

customername	ordernumber	productname
Atelier graphique	10123	1965 Aston Martin DB5
Atelier graphique	10123	1999 Indy 500 Monte Carlo SS
Atelier graphique	10123	1948 Porsche Type 356 Roadster
Atelier graphique	10123	1966 Shelby Cobra 427 S/C
Atelier graphique	10298	1996 Moto Guzzi 1100i
Atelier graphique	10298	1936 Harley Davidson El Knucklehead
Atelier graphique	10345	1938 Cadillac V-16 Presidential Limou...
Signal Gift Stores	10124	1917 Grand Touring Sedan
Signal Gift Stores	10124	1911 Ford Town Car
Signal Gift Stores	10124	1932 Model A Ford J-Coupe
Signal Gift Stores	10124	1932 Alfa Romeo 8C2300 Spider Sport
Signal Gift Stores	10124	1957 Ford Thunderbird
Signal Gift Stores	10124	1970 Chevy Chevelle SS 454
Signal Gift Stores	10124	1939 Chevrolet Deluxe Coupe
Signal Gift Stores	10124	1938 Cadillac V-16 Presidential Limou...
Signal Gift Stores	10124	1949 Jaguar XK 120
Signal Gift Stores	10124	1952 Citroen-15CV
Signal Gift Stores	10124	1969 Chevrolet Camaro Z28
Signal Gift Stores	10124	2002 Chevy Corvette
Signal Gift Stores	10124	1936 Mercedes Benz 500k Roadster
Signal Gift Stores	10278	1993 Mazda RX-7
Signal Gift Stores	10278	1965 Aston Martin DB5

20. Find customers who made payments greater than 100000.

select customername,amount from customers

inner join payments

using(customernumber)

where amount > 100000;

output:

customername	amount
Mini Gifts Distributors Ltd.	101244.59
Mini Gifts Distributors Ltd.	111654.40
Euro+ Shopping Channel	116208.40
Euro+ Shopping Channel	120166.58
Dragon Souvenirs, Ltd.	105743.00

21.Find the customer who generated the highest revenue.

select customername, sum(quantityordered * priceeach) as total_sales

from customers

inner join orders

using(customernumber)

inner join orderdetails

using(ordernumber)

group by customername

order by total_sales desc

limit 1;

output:

customername	total
Euro+ Shopping Channel	820689.54

22. Find the most ordered product.

```
select productname, sum(quantityordered) as total_quantity
from products
inner join orderdetails
using(productcode)
group by productname
order by total_quantity desc
limit 1;
```

output:

productname	total
1992 Ferrari 360 Spider red	1808

23. Find the country generating the highest sales.

```
select country, sum(quantityordered * priceeach) as total_sales
from customers
inner join orders
using(customernumber)
inner join orderdetails
using(ordernumber)
group by country
order by total_sales desc
limit 1;
```

output:

country	sales
USA	3273280.05

24.Find customers with payments greater than the average payment.

select customername,amount from customers

inner join payments

using(customernumber)

where amount >

(select avg(amount) from payments)

order by amount desc;

output:

customername	amount
Euro+ Shopping Channel	120166.58
Euro+ Shopping Channel	116208.40
Mini Gifts Distributors Ltd.	111654.40
Dragon Souvenirs, Ltd.	105743.00
Mini Gifts Distributors Ltd.	101244.59
Corporate Gift Ideas Co.	85559.12
Mini Gifts Distributors Ltd.	85410.87
Herkku Gifts	85024.46
Mini Gifts Distributors Ltd.	83598.04
Australian Collectors, Co.	82261.22
Collectable Mini Designs Co.	80375.24
Down Under Souvenirs, ...	75020.13
Euro+ Shopping Channel	65071.26
Euro+ Shopping Channel	63843.55
Diecast Classics Inc.	63357.13
Vida Sport, Ltd	61402.00
Toms Spezialitäten, Ltd	61234.67
Euro+ Shopping Channel	59830.55
The Sharp Gifts Warehouse	59551.38
Diecast Collectables	59265.14
Muscle Machine Inc	58841.35
Muscle Machine Inc	58793.53

25. Find monthly sales trend.

SELECT

YEAR(orderdate) AS year,

MONTH(orderdate) AS month,

SUM(quantityordered * priceeach) AS total_sales

FROM orders

JOIN orderdetails USING(ordernumber)

GROUP BY YEAR(orderdate), MONTH(orderdate)

ORDER BY YEAR(orderdate), MONTH(orderdate);

output:

month_sal	sales
1	716815.00
2	735098.65
3	737920.36
4	718244.98
5	869235.79
6	493841.51
7	527503.85
8	597584.20
9	520497.65
10	1014570.07
11	1967317.13
12	705561.42

Key Insights

- A small number of customers contribute a large portion of total revenue (high-value customers).
- Certain products (like top-selling items) generate the majority of sales.
- The country with the highest sales contributes significantly to overall revenue.
- Monthly sales analysis shows variation in revenue across different months, helping identify peak sales periods.
- Some customers have not placed any orders, indicating potential targets for marketing campaigns.
- Customers with higher credit limits tend to generate more revenue.
- Identifying top customers helps businesses focus on customer retention strategies.

Skills Demonstrated:

- SQL Joins
- Group By
- Aggregate Functions
- Subqueries
- Data Analysis
- Business Insight Generation

Conclusion

In this project, SQL was used to analyze customer, order, and sales data from the ClassicModels database. The analysis helped identify top customers, best-selling products, and high-revenue generating countries. Monthly sales trend analysis helped understand sales performance over time. The project demonstrates the use of SQL joins, aggregate functions, group by, and subqueries to extract business insights and support data-driven decision-making.

About This Project

This project was created as part of my Data Analyst portfolio to demonstrate SQL skills, data analysis, and business insight generation using the ClassicModels database. The project focuses on analyzing sales performance, customer behavior, and product performance using MySQL.

Thank you for reviewing my SQL project.

Tools Used: MySQL

Created by: Akshada Kumbhar